NAB2 → Th1/Th2 → atopic eczema, hop by hop

HOP 0 Knockdown QC
experimental receipt

2/2 guides significant
best adj. $p \approx 1 \times 10^{-16}$
guide 0.056 vs NTC 0.567

The perturbation worked.

HOP 1 Effect
experimental receipt

on-target knockdown
effect −16.9
301 downstream DE genes

Large, clean effect.

HOP 2 Program
experimental receipt

Th2 in Ota contrast
$z = 7.71$, adj. $p = 1.96 \times 10^{-13}$
log-FC +0.63

Shifts the T-cell program.

HOP 3 Disease
association receipt

eczema modules 90 & 100
OR 3.90 (FDR 0.0028)
OR 3.43 (FDR 0.0224)

genetic-association nomination

VERDICT: supported -- a receipt-backed regulatory nomination

STAT6 cis-effect: RULED OUT (S4.4)

Under NAB2-KD (authors' genome-wide DE):
STAT6 $\log_2$FC +0.087, adj. p 0.788 → UNMOVED
(ranks 5,444 / 10,282). NAB2 self −3.078.
A cis-artifact would push STAT6 down;
it does not move -- the signal is NAB2-specific.

12q13 caveat (S4.4b)

The eczema label is a GWAS association with no colocalization / LD control. NAB2 and STAT6 are convergent neighbors (3′ ends overlap; promoters ~43 kb apart) in the 12q13 atopy locus, so the disease-label provenance is OPEN -- foregrounded, not claimed discharged.

Two receipt classes, kept distinct

Hops 0-2 (blue) = EXPERIMENTAL receipts (direct perturbation measurements). They license the claim 'consistent with a re-derived NAB2 -> Th1/Th2 regulatory role the literature has not made.'

Hop 3 (purple) = ASSOCIATION receipt (a GWAS-based disease label). Licenses a NOMINATION, not a claim of experimental disease causality. Independent 4-agent audit: 0 papers link NAB2 to either node.